

KATHIRVEL A

EMBEDDED DEVELOPER



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[LinkedIn Profile](#)



[GitHub](#)



[Portfolio](#)



[Karur, TamilNadu](#)

CAREER OBJECTIVES

I am an Embedded systems enthusiast with strong hands-on experience in STM32, ESP32 and bare-metal Embedded C. Seeking roles in product development and Embedded systems engineering .

SKILLS

- Languages: C, Embedded C, C++, Python
- Embedded Concepts: Bare-Metal Programming
- Communication Protocols: UART/USART, SPI, I2C
- MCU Peripherals: ADC, DAC, Timers, PWM
- Microcontrollers: STM32 (ARM Cortex-M), ESP32, Arduino
- PCB Design: Basic schematic and layout understanding
- Tools: STM32CubeIDE, Arduino IDE, VS Code, KiCad 9.0, Fusion 360
- Computer Vision: Hand Detection, Pose Detection, Face Detection (MediaPipe)

INTERSHIP

- **Elamani Tech Pvt Ltd**

Disassembled and analyzed a commercial service robot to study its internal hardware architecture, including motors, encoders, sensors, control boards, and system-level wiring.

PROJECTS

- **DataLogger Using STM32:**

Developed an STM32-based data logger to acquire GPS (USART) and analog sensor (ADC) data with interrupt-driven processing. Stored data in W25Q64 SPI flash and enabled real-time display via UART communication.

- **AI Vision-based Wireless Robot:**

Built a computer vision-based robot that detects real-time hand gestures using MediaPipe. Integrated Arduino with Bluetooth communication to enable wireless gesture-controlled movement.

- **ExpiraTrack – Smart Expiry Monitoring Website**

Developed an AI-powered expiry tracking system using Generative AI tools for intelligent date monitoring. Designed to automate alerts and improve inventory efficiency with smart tracking logic.

EDUCATION

- HSE-93.4%
- Electronics and Communication Engineering
Government College of Engineering, Salem (2023-2027)
CGPA-7.72

LEADERSHIP & ACTIVITIES

- Served as class representative from the second semester to the present, acting as a liaison between students and faculty and coordinating academic communication and class activities.